

QUAZI RIHAL MAHMOOD

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Professional Summary

Computer Science undergraduate with a strong foundation in **full-stack web development (MERN stack)** and **mobile application development**. Experienced in building **scalable web and mobile applications**, **RESTful APIs**, and handling **data-driven systems**. Skilled in **JavaScript**, **TypeScript**, **backend architecture**, and **database design**, with a growing interest in developing efficient, user-focused, and high-performance applications.

Technical Skills

Languages: JavaScript, TypeScript, Python, C, C++, Java, SQL
Frontend: React.js, HTML, CSS
Backend: Node.js, Express.js
Databases: MongoDB, MySQL
Tools: Git, Postman, Docker

Experience

Teaching Assistant

North South University

Dhaka, Bangladesh

July 2024 – August 2025

- Served as TA for five courses: **C Programming**, **Java**, **Database Systems**, **Digital Logic Design**, and **Computer Architecture**.
- Evaluated assignments, quizzes, and exams across multiple courses, providing constructive feedback to improve student performance.
- Mentored undergraduate students on problem-solving approaches spanning both software and hardware domains.

Projects

Colossus (MERN Stack) | MongoDB, Express.js, React.js, Node.js, Google Drive API, OAuth 2.0

- Designed and built a full-stack cloud storage platform that aggregates multiple Google Drive accounts into a unified storage system.
- Implemented an **auto-chunking algorithm** to split large files across multiple drives and reassemble them seamlessly during download.
- Developed **RESTful APIs** for file upload, retrieval, deletion, and drive management using Node.js and Express.
- Integrated **Google OAuth 2.0** for secure third-party authentication and drive access.
- Designed scalable **MongoDB schemas** to store file metadata and chunk mapping for distributed storage.
- Built a responsive **React** frontend enabling file management, search, and storage visualization.
- Developed a **companion mobile app in Java**, enabling users to upload, access, and manage files on the go.

DraftSense | MongoDB, Express.js, React.js, Node.js, TypeScript, Ollama, Riot Data Dragon API

- Engineered a full-stack **AI-powered** League of Legends champion draft advisor using the MERN stack with TypeScript, enabling real-time pick recommendations during ranked match champion select.
- Designed and implemented a **dual-scoring recommendation engine** computing counter scores (candidate vs. enemy winrates) and synergy scores (candidate with ally winrates), surfacing the top 5 counter picks, 5 synergy picks, and 3 overall best picks per draft state.
- Integrated **Ollama with llama3.1:8b** as a local LLM backend to generate natural language explanations for each recommended champion, running fully on-device without any external API dependency.
- Managed global draft state across 4 pages using **Zustand**, seeded champion and matchup data from Riot Games' Data Dragon API, and secured the Express backend with **Helmet** and **rate limiting**.

Education

North South University

Bachelor of Computer Science Engineering | CGPA: 3.72/4.00

Dhaka, Bangladesh

Expected Graduation: May 2026

Reference

Dr. Mohammad Rashedur Rahman

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North South University
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